# QUALITY ASSURANCE TESTING DOCUMENTATION

# (QATD)

## UMHackathon 2026

## Four Fingers

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# Document Control

| Field | Detail |
|---|---|
| **System Under Test (SUT)** | UMHackathon 2026 - Team **Four Fingers** |
| **Team Repo URL** | https://github.com/qiphy/talentflow |
| **Project Board URL** | https://tinyurl.com/3bfe6tth |
| **Live Deployment URL (optional for preliminary round)** | - |

# Objective:

The primary objective is ensuring TalentFlow can reliably handle candidate application submissions, processing onboarding to be monitored whilst providing up to date logs of the GLM activities.

# PRELIMINARY ROUND (Test Strategy & Planning)

## 1. Scope & Requirements Traceability

This section assures that all implemented features of TalentFlow are properly validated through repeated testing. A Requirement Traceability Matrix links the functioning requirements to appropriately simulated test cases to ensure the platform operates as intended.

### 1.1 In-Scope Core Features

- **Candidate Application Form:** Candidates can upload their resumes/CV for parsing the required information, or manually fill in their application details.
- **Application Monitoring:** Provide candidates a viewable list of their previous submissions with GLM generated feedback, additionally feedback provided from the employer.
- **Employee Monitoring Dashboard:** Dashboard with organised details of all candidate applications for employer viewing.
- **Activity Log Monitoring:** Operational logs of all user activities are updated in a list for monitoring
- **AI-Assisted Monitoring:** Z AI GLM generates automated summaries of all activity logs

### 1.2 Out-of-Scope

- Full HR Management System: No payroll management or employee lifecycle included
- Advanced Ai Decision Making: AI maintains zero control over any actual hiring

## 2. Risk Assessment & Mitigation Strategy

*The technical risks below are identified from TalentFlow's application requirements and architecture using the 5x5 Risk Assessment Matrix.*

**2.1 Risk Assessment Scoring Criteria:**

| *Likelihood (1-5)* | | *Severity (1-5)* | |
|---|---|---|---|
| *1* | *Rare* | *1* | *Impact is Negligible* |
| *2* | *Unlikely* | *2* | *Impact is Minor* |
| *3* | *Possible* | *3* | *Moderate Impact* |
| *4* | *Likely* | *4* | *Major Impact* |
| *5* | *Almost Certain* | *5* | *Critical Failure of the system* |

***Risk Score = Likelihood × Severity***

**2.2 Risk Level Reference:**

| ***Risk Score*** | ***Risk Level*** | ***Recommended Action*** |
|---|---|---|
| *1 – 5* | *Low* | *Monitor only. Acceptable risks.* |
| *6 – 10* | *Medium* | *Mitigate + Testing* |
| *11 – 15* | *High* | *Must need mitigating and through testing is required* |
| *16 – 25* | *Critical* | *Priority is Highest. Need extensive level of testing.* |

**2.3 Technical Risk Register:**

| **Technical Risk** | **Likelihood (1–5)** | **Severity (1–5)** | **Risk Score (L×S)** | **Mitigation Strategy** | **Testing Approach** |
|---|---|---|---|---|---|
| AI analysis unavailable or returns malformed JSON on application submission. | 3 | 3 | 9 (medium) | Implement automatic rollback strategy to handle such failures. [e.g. Database transaction timeout increased to 10s] | Force a bad API key locally and submit an application. Verification saves without crashing and surfaces a graceful error to the user. |
| Session cookie missing or expired mid-session | 4 | 3 | 12 (high) | Return HTTP 401 from all protected routes; frontend redirects to login. | Manually deletes the access_token cookie in DevTools and attempts a protected action. The user is redirected to the login page. |
| PDF CV upload fails or contains no | 3 | 2 | 6 (medium) | Validate extracted text length before sending to AI. If the | A scanned (imaged-only) PDF is uploaded and the |

| Technical Risk | Likelihood (1–5) | Severity (1–5) | Risk Score (L×S) | Mitigation Strategy | Testing Approach |
|---|---|---|---|---|---|
| extractable text | | | | file is empty, HTTP 400 with descriptive error is empty and requests the user to upload a text-based PDF. | HTTP 400 error is returned and displayed on the frontend. |
| Employer role verification bypassed | 2 | 5 | 10 (medium) | Role check is enforced server-side on every protected route via get_current_user + profile lookup. Front-end redirect alone are not as reliable. | Log in as an employee, manually navigate it to /employerHome URL to confirm server returns 403 and page is inaccessible. |
| Company mismatch check bypassed, causing employer to log in under a different company's account. | 2 | 4 | 8 (medium) | Case-insensitive, strip-whitespace company comparison enforced at login. Mismatch returns HTTP 401 and signs the user out of Supabase Auth. | Attempt login with correct credentials but a mismatched company name — confirm 401 is returned and no session cookie is set. |
| Supabase DB write fails silently. Application is submitted but not stored. | 2 | 5 | 10 (medium) | Check response from supabase.table().insert().execute(); raise HTTPException on failure. Add structured logging for all DB write errors. | Simulate a DB failure (e.g. wrong table name in staging) and verify HTTP 500 is returned and the user is not shown a false success. |
| CORS misconfiguration causes frontend requests in production to be blocked. | 3 | 4 | 12 (high) | CORS origins are explicitly whitelisted in main.py. Before deployment, update allow_origins to the production domain and test all fetch calls. | Deploy to a test URL not in the whitelist and confirm CORS errors appear. Then add the URL and confirm requests succeed. |
| OpenRouter / ilmu.ai API rate limit hit during | 3 | 3 | 9 (medium) | Catch API rate-limit exceptions; fall back to null AI score with a logged warning. | Submit multiple applications in rapid succession and verify each save |

| Technical Risk | Likelihood (1–5) | Severity (1–5) | Risk Score (L×S) | Mitigation Strategy | Testing Approach |
|---|---|---|---|---|---|
| high-volume application submission. | | | | Consider queuing AI analysis as a background task. | correctly even if AI analysis degrades or fails. |
| Sensitive credentials in .env committed to version control. | 2 | 5 | 10 (medium) | Add .env to .gitignore before the first commit. Use environment variable injection in production. Rotate any exposed keys immediately. | Run git status and git log -- .env to confirm the file is not tracked. Audit commit history for any previously committed secrets. |
| Background log_ai_event call fails and raises unhandled exception. | 3 | 2 | 6 (medium) | Wrap log_ai_event body in try, failure should log to console but never propagate to the user-facing response. | Temporarily break the activity_logs table schema and submit an application and confirm the main response still returns 200 and the error is only logged server-side. |

## 3. Test Environment & Execution Strategy

*Explain the testing environment. That means where the testing of your application will take place, how you are going to handle test data and the rules/threshold that you are incorporating to govern your testing phases.*

- **Unit Test**
    - Scope: Candidate intake validation, submission handling, monitoring dashboard data, activity log generating
    - Execution: Tests are written whenever new features are implemented. Execution proceeds during development including throughout the CI pipeline once pushed into the repository.
    - Isolation: In ensuring accuracy of internal logic, external dependencies such as database connections and AI responses are observed using placeholder data.
    - Pass Condition: Returned output is under expectations (i.e Valid candidate submissions, invalid form inputs & accounting for edge cases with empty fields or unexpected values)
- **Integration Test**
    - Scope: Candidate applications go to the local storage, then to the monitoring dashboard for display. All activities should be moved to be generated as a list of activity logs
    - Execution: Integration tests are performed once each individual component functions successfully and are merged into the main branch.
    - Workflow: Ensure real database queries are executed and the API endpoints are called. The monitoring pages should retrieve the stored submissions.
    - Pass Condition: When a candidate submits a successful application, the monitoring page should immediately display said submission once retrieved from the database.
- **Test Environment (CI/CD Practice):**
    - Local Testing: Manual testing using own device as a local host
    - CI: Automated tests are triggered on every push of code into the repo.
    - Automated CI/CD pipeline: Perform automated checks for successful execution and verification of code compilation. Upon any failure, the pipeline prevents merging with the main branch.
- **Regression Testing & Pass/Fail Rules:**
    - Execution Phase: Regression testing happens when major updates are introduced and branches are merged into the main development branch.
    - Pass/Fail Condition: The test case passes when the system behaviour matches the expected result.
    - Continuation Rule: Critical system tests for candidate submission and monitoring information retrieval are prioritised. Those tests must pass before continuing through other testing phases.
- **Test Data Strategy:**
    - Manual: Test accounts are created to simulate the two primary roles. In which candidate users can submit applications and employers having viewable access to all monitoring logs.
    - Automated: Database seed scripts generate sample candidate submissions and activity logs automatically. Allowing for the system to be tested under similar conditions to a live scenario.
- **Passing Rate Threshold:**

- At least 85% of test cases must pass before the code can be deployed. System tests in regards to candidate submissions and monitoring must be 100% successful whilst AI-assisted outputs have to be achieve a minimum of 80% acceptable response rate

## 4. CI/CD Release Thresholds & Automation Gates

*This section is mainly for defining metrics & threshold for success criteria. So, if a certain threshold is not met, the pipeline blocks the forward transition.*

### 4.1 Integration Thresholds (Merging to Main)

Ensuring the main branch is stable for development.

| Check | Requirements | Project | Pass/Failed |
|---|---|---|---|
| Automatic Build | Zero build errors | 0 Errors | Passed |
| Unit Tests | Minimal 95% passing rate | 100% | Passed |
| Form Validation Test | All intake validation tests must pass | 100% | Passed |
| Code Quality | No critical linting errors | 94% | Passed |
| Test Coverage | Minimum 80% coverage | 82% | Passed |

### 4.2 Deployment Thresholds ( Production)

Ensuring system is stable for release

| Checks | Requirements | Project | Pass/Failed |
|---|---|---|---|
| Regression Tests | Minimum 90% pass rate | 95% | Passed |
| Monitoring API Performance | Average response time < 800ms | 550ms | Passed |
| Database Integrity | Form submissions stored successfully | Verified | Passed |
| Critical Bugs | Zero P0/01 bugs | 0 | Passed |
| Security | Zero exposed API keys or credentials | Clean | Passed |

## 5. Test Case Specifications (Drafts)

| Test Case ID | Test Type & Mapped Feature | Test Description | Test Steps | Expected Result | Actual Result |
|---|---|---|---|---|---|
| TC-01 | **Happy Case (Entire Flow):** Candidate Submission Flow | Verify for a candidate to submit an application and viewing the application log in the Monitor Applications page. | 1. Open TalentFlow, login as an employee.<br>2. Head to New Intake page<br>3. Fill up the candidate intake form<br>4. Submit the application<br>5. Open the Monitor Applications page to view the application. | The candidate application is submitted and successfully stored in the system database. It should appear in the Monitor Application page as a log. | Application is submitted and displayed on the Monitor Application page.<br><br>Status: Passed |
| TC-02 | **Specific Case (Negative/Edge Test):** Empty Required Fields | Verify if the system rejects submission when there exists required fields that are missing. | 1. Open the New Intake page as an employee.<br>2. Fill up the submission form while leaving some required fields blank.<br>3. Attempt to submit form | The system prevents the form being submitted. Required fields to be filled are highlighted in red | Submission is blocked, with the empty required fields highlighted in red to notify candidates.<br><br>Status: Passed |
| TC-03 | **NFR (Performance):** Stress Testing Large Input | Check capabilities of form by pushing large sized input data into the form field. | 1. Open the New Intake page<br>2. Fill out the application form<br>3. Enter in a long input for one of the fields<br>4. Submit the form | System truncates without any UI crash | The system became unresponsive. The submission still goes through and is logged in the database.<br><br>Status: Failed |
| TC-04 | **NFR (Performance):** Backend API Performance | Verify the resume AI extraction is operating at an acceptable rate | 1. Open any browser developer tools on TalentFlow<br>2. Navigate to the New Intake page<br>3. Submit a resume for parsing and observe upload time.<br>4. Repeat process with different resumes and calculate the average | API response time should ideally be between 800-1000 ms. No server errors should occur either. | AI parsing fails with resumes above 100KB. The average response time across 4 resumes is 1 minute 14 seconds 560 ms.<br><br>Status : Failed |

## 6. AI Output & Boundary Testing (Drafts)

This section is designed to validate that your GLM integration does produce acceptable output and handles any inputs that are abnormal gracefully.

6.1. **Prompt/Response Test Pairs**

| Test ID | Prompt Input | Expected Output (Acceptance Criteria) | Actual Output | Status |
|---|---|---|---|---|
| AI-01 | Personalised resume/CV to be parsed containing candidate information | All appropriate fields should be automatically filled. | Each field specified in the resume is filled accurately according to the candidate resume. | Pass |
| AI-02 | Personalised resume/CV to be parsed contains a long list of candidate skills | Form should list all skills listed in the resume/CV | All skills identified in resume is automatically added in the Work Experiences and Key Skills field | Pass |
| AI-03 | Analyse candidate submission with missing fields | AI should identify absence of required information | AI prevents submission of form till all required fields are filled | Pass |
| AI-04 | Incorrect file format (JPG of resume) is submitted (i.e System only accepts PDF or DOCX) | Reject the file upload, indicating Ai extracting is unsuccessful | Pop-up message appeared saying that the AI extraction failed. No information from that resume was parsed into the form | Pass |

**6.2. Oversized/Larger Input Test**

| Fields | Details |
|---|---|
| Maximum Resume Upload Size | 100KB |
| Upload size used while testing | 87KB, 103KB, 43KB, 324KB |
| Expected Behavior | The AI parsing fails for resumes more than 100KB |
| Actual Behavior | The 43KB and 87KB sized resumes are successfully extracted whilst the 103KB and 324KB sized resumes are rejected |
| Status | Pass |

| Fields | Details |
|---|---|
| Maximum Text Input Size | Not explicitly stated |
| Input used while testing | 200+ characters in the 'Work Experiences and Key Skills' field |
| Expected Behavior | Response time for applicant submission is delayed |
| Actual Behavior | Submission system becomes completely unresponsive |
| Status | Failed |

**6.3. Adversarial/Edge Prompt Test**

Test Scenario :
The candidate submission should be consistently in line with candidate information, implying that no tolerance is given to unserious submissions. In this instance, a candidate applying enters in an invalid skill in the 'Work Experiences and Key Skills' field i.e "Being Really Good".

Expected Behaviour :

- The input is registered and added to the form but should be flagged upon form submission by the system.

Actual Behaviour :

- The form is submitted with non credible skills listed under the candidate's resume. No filtering or normalization of the field occurs before processing

Status : Failed

**6.4. Hallucinating Handling**

Hallucinations in responses occur in submitting resumes/CV that are exaggerated or unclear in defining field criteria. With AI inferring inaccurate job experiences that are not typically acceptable under company hiring quota.

Currently, TalentFlow does not have a dedicated hallucination detection model. It relies on standardized input that is entered by either the user or AI GLM. Upon submission, the system does not detect it and includes the raw data as is into the database. This feature is only planned as a future improvement.

***Important Note:***

*Using Agile principle, these testing will take place iteratively over the period of development, not limited to at the end.*